Notes: Chen, Jiang, Liu, Suárez Serrato & Xu (RES, 2023)

Tax Policy and Lumpy Investment Behaviour: Evidence from China's VAT Reform

Contents

1 Big picture

- **Question.** How does tax policy affect investment when investment is *lumpy* rather than smooth? The paper asks whether tax reforms operate not only through the usual user-cost channel, but also through the *extensive margin*: whether firms invest at all.
- **Setting.** China's 2009 nationwide reform of the value-added tax on equipment investment. Before the reform, most domestic firms paid VAT on equipment and could not deduct it; after the reform they could deduct input VAT on new equipment.
- **Core economic idea.** A tax on investment raises the purchase price of capital but not the resale price of used capital, so it creates *partial irreversibility*. In a model with fixed and convex adjustment costs, that widens the inaction region and suppresses investment spikes.
- **Identification.** Domestic firms are the treated group. The control group is foreign firms in encouraged industries that were already allowed to deduct VAT on equipment before 2009. The reform therefore changed domestic firms' tax treatment but not these foreign firms'.
- **Main reduced-form result.** Relative to foreign firms, domestic firms become about 5 percentage points more likely to invest and raise the equipment-investment rate by about 3.6 percentage points of capital, which is a 36% increase relative to a baseline investment rate of 10%.
- **Main mechanism result.** The reform mainly works through *investment spikes*. The spike rate rises by about 7.3 percentage points, and spike investments explain about 86% to 92% of the increase in the investment rate.
- **Structural result.** A dynamic model with convex adjustment costs, fixed disruption costs, and partial irreversibility matches both pre-reform lumpy-investment moments and the reduced-form effects of the reform. The preferred estimates imply $\gamma \approx 1.434$, an upper bound of fixed costs $\bar{\xi} \approx 0.118$, and economic depreciation $\delta \approx 0.071$.
- **Policy conclusion.** Policies that directly shrink the inaction region—such as eliminating the VAT distortion or granting an investment tax credit—are more cost-effective at stimulating investment than policies that merely lower the user cost, such as a corporate income tax cut.
- **Why the paper matters.** The paper shows that the usual user-cost formula is not a sufficient statistic in environments with lumpy investment. The design of tax policy matters because different tax instruments interact differently with the frictions that generate inaction and spikes.

2 Data and measurement (institutional setting and key objects)

2.1 The 2009 Chinese VAT reform

- **Pre-reform system.** China's pre-2009 "production-based" VAT let firms offset output VAT with VAT on intermediate inputs, but not with VAT paid on fixed investment. Equipment purchases therefore carried a tax wedge.
- **Reform path.** The move to a "consumption-based" VAT started with pilot programs in 2004 in Northeastern provinces, then expanded nationwide unexpectedly on 1 January 2009 after a surprise announcement on 19 December 2008.

- **Why it is a large policy shock.** The reform lowered the after-tax cost of investment by close to 15%, much larger than the roughly 4% user-cost reduction often attributed to the recent U.S. tax reform.
- **Control group logic.** Foreign firms in encouraged industries had already been allowed to deduct equipment purchases from VAT, so the national rollout did not change their tax treatment.
- **Unexpected and permanent.** The surprise timing reduces concerns about anticipation. The permanence reduces concerns that the results are short-run intertemporal shifting only.

2.2 A simple tax-cost calculation: Table 1

- For a 1,000 RMB equipment purchase, the pre-reform VAT-inclusive cost is 1,170 RMB.
- Before the reform, no VAT credit is available, book value is 1,170, the present value of depreciation deductions is 948.6, and corporate-tax deductions are worth 237.2. The after-tax cost is therefore 932.8 RMB.
- After the reform, the firm can deduct 170 RMB of VAT immediately, but book value falls to 1,000, so future depreciation deductions are smaller: the present value of depreciation is 810.8 and the associated corporate-tax saving is 202.7.
- Net effect: the after-tax cost falls from 932.8 to 797.3, a drop of 135.5 RMB, or roughly 15%.

2.3 Main data

- **Primary source.** National Tax Survey Database from China's State Administration of Tax, 2007–2011.
- **Sample design.** Large firms appear every year; smaller firms rotate in and out. The paper restricts to firms with non-negative productive fixed assets and stable ownership type.
- **What is observed.** Equipment investment, investment in structures, fixed assets, sales, cash inflow, debt, tax liabilities, and pilot-program participation.
- **Additional matches.** Ministry of Commerce foreign-investment records identify which foreign firms had preferential treatment; the Annual Survey of Manufacturing (2005–2006) extends pre-trends backward.
- **Scale.** The raw tax sample contains about 315,000 firm-year observations. The main balanced-panel regressions use about 86,870 observations for the extensive-margin outcome and about 81,270 for the investment-rate outcome.

2.4 Summary statistics and lumpy-investment facts

- Average equipment investment is 3.15 million RMB overall, 2.61 for domestic firms, and 9.01 for foreign firms.
- The mean equipment-investment rate is 0.10 for both domestic and foreign firms, measured as equipment investment divided by the stock of net fixed assets.

- Investment is highly lumpy: 49% of firms do not invest in a given year, and roughly 17% replace more than 20% of their capital stock.
- The one-year autocorrelation of the investment rate is around 0.20, which suggests that the data need both nonconvex frictions (to generate inaction and spikes) and convex frictions (to generate sluggish responses).
- Domestic firms are smaller than foreign firms on average in sales, assets, cash flow, and debt, so the paper checks robustness with inverse-probability reweighting to match observables across groups.

3 Models and empirical specifications (all core equations + interpretation)

3.1 Frictionless benchmark and the user cost of capital

The static benchmark firm solves

$$\max_{K} \, (1-\tau)A^{1-\theta}K^{\theta} - p(K-K_0) \quad \Longrightarrow \quad K^* = A\left[\frac{1}{\theta}\frac{p_k(1+\nu)(1-\tau z)}{1-\tau}\right]^{-\frac{1}{1-\theta}}. \tag{1}$$

The tax-inclusive user cost of capital is

$$\text{UCC} = \frac{p_k(1+\nu)(1-\tau z)}{1-\tau}.$$

- **Interpretation.** Without investment frictions, all tax reforms only matter through the user cost. A fall in ν, a rise in z, or a fall in τ changes K^* smoothly; there is no inaction region and no extensive-margin response.
- **Elasticity logic.** In this benchmark, the investment elasticity with respect to the user cost is $-1/(1-\theta)$, so the public-finance tradition of estimating a user-cost elasticity is natural *only* in the frictionless case.

3.2 Partial irreversibility and the inaction region

With a purchase price p_b and resale price p_s such that $p_b > p_s$, the inaction region is

$$[A, \bar{A}] = \left[\left(\frac{p_s}{\theta(1-\tau)}\right)^{\frac{1}{1-\theta}} K_0,\ \left(\frac{p_b}{\theta(1-\tau)}\right)^{\frac{1}{1-\theta}} K_0\right]. \tag{2}$$

Using the Chinese tax system, the paper writes the width of the inaction region as

$$\bar{A} - A = \left(\frac{\text{UCC}}{\theta}\right)^{\frac{1}{1-\theta}} \left[1 - \left(\frac{1-\tau(1+\nu)z}{(1+\nu)(1-\tau z)}\right)^{\frac{1}{1-\theta}}\right] K_0. \tag{3}$$

- **Interpretation.** A higher VAT wedge ν makes the purchase price of new capital high relative to the resale price of used capital, so firms tolerate a wider range of productivity realizations without adjusting.

- **Economic lesson.** A VAT cut does two things at once: it lowers the user cost *and* narrows the purchase-resale price gap. Both push firms out of inaction and into investment.
- **Why this is different from bonus depreciation.** Raising z lowers the user cost but can widen the effective price wedge, so policies with the same user-cost change can have different extensive-margin effects.

3.3 Fixed disruption costs

With a non-deductible fixed cost equal to a fraction ξ of desired capital, the adjustment/no-adjustment indifference condition is

$$\left[\frac{1-\theta}{\theta}\mathrm{UCC} - \frac{\xi}{1-\tau}\right]\left(\frac{\theta}{\mathrm{UCC}}\right)^{\frac{1}{1-\theta}} A + \mathrm{UCC}K_0 = K_0^{\theta}A^{1-\theta}. \tag{4}$$

- **Interpretation.** Fixed costs generate another source of inaction. A firm only adjusts when the gain from moving to the optimal capital stock is big enough to cover the disruption cost.
- **Tax-policy contrast.** A CIT cut lowers the user cost *and* lowers the after-tax value of the fixed adjustment cost, so it interacts differently with lumpiness than a VAT reform.
- **Main conceptual point.** The user cost is no longer a sufficient statistic once tax policy changes the frictions themselves.

3.4 Dynamic investment model

The period-profit function is

$$\Pi(K, A_{\Pi}) = (A_{\Pi})^{1-\theta}K^{\theta}, \tag{5}$$

and log profitability is decomposed as

$$a_{it} = b_t + \omega_i + \varepsilon_{it}, \tag{6}$$

where b_t is an aggregate shock, ω_i is permanent firm heterogeneity, and ε_{it} is a transitory AR(1) shock.

The normalized value function is

$$v(k, b, \varepsilon, \xi) = \max\{v^b(k, b, \varepsilon, \xi),\, v^s(k, b, \varepsilon, \xi),\, v^i(k, b, \varepsilon, \xi)\}, \tag{7}$$

with

$$v^b(k,b,\varepsilon,\xi) = \max_{i>0} \left\{ (1-\tau)\pi(k,b,\varepsilon) - \left([1+\nu-\tau z(1+\nu)]i + \frac{\gamma}{2}\left(\frac{i}{k}\right)^2 k + \xi k^* \right) + \beta E\big[v^0(k',b',\varepsilon') \mid b,\varepsilon\big] \right\}, \tag{8}$$

$$v^s(k,b,\varepsilon,\xi) = \max_{i<0} \left\{ (1-\tau)\pi(k,b,\varepsilon) - \left([1-\tau z(1+\nu)]i + \frac{\gamma}{2}\left(\frac{i}{k}\right)^2 k + \xi k^* \right) + \beta E\big[v^0(k',b',\varepsilon') \mid b,\varepsilon\big] \right\}, \tag{9}$$

$$v^i(k,b,\varepsilon,\xi) = (1-\tau)\pi(k,b,\varepsilon) + \beta E\big[v^0(k(1-\delta),b',\varepsilon') \mid b,\varepsilon\big], \tag{10}$$

where $k' = (1-\delta)k + i$.

- **Frictions in the model.** Convex adjustment costs $\frac{\gamma}{2}(i/k)^2 k$, fixed disruption costs ξk^*, and partial irreversibility because the purchase price includes VAT while the resale price does not.
- **Time-to-build.** Current-period investment does not raise current profits; it only affects next period's capital stock.
- **Comparative statics.** When the paper solves the model numerically, a VAT cut shrinks the inaction region much more strongly than an equivalent user-cost change via a CIT cut or bonus depreciation.

3.5 Tax component of the user cost

The paper summarizes the tax component of the user cost as

$$\text{TUCC} = (1+\nu)\frac{1-\tau z}{1-\tau}. \tag{11}$$

- **Interpretation.** The VAT increases the purchase price directly through $(1+\nu)$ and indirectly through the book value that determines depreciation deductions.
- **Quantitative comparison.** The 2009 reform lowers TUCC by about 15%. Setting $z = 1$ via expensing lowers TUCC by only about 6%, and eliminating the income tax also lowers TUCC by only about 6%.
- **Lesson.** Even when TUCC changes are comparable, the overall investment effect need not be, because the instruments have different effects on partial irreversibility and fixed costs.

3.6 Reduced-form event study and difference-in-differences

The event-study specification is

$$Y_{ijt} = G_i\gamma_t + \mu_i + \delta_{jt} + X'_{it}\beta + \varepsilon_{ijt}, \tag{12}$$

where $G_i = 1$ for domestic firms and 0 for the foreign control group.

The main difference-in-differences specification is

$$Y_{ijt} = \gamma\, G_i \times Post_t + \mu_i + \delta_{jt} + X_{it}'\beta + \varepsilon_{ijt}. \tag{13}$$

- **Outcomes.** The paper studies both the extensive margin (a dummy for positive equipment investment) and the intensive margin (equipment investment divided by fixed assets).
- **Fixed effects.** Firm fixed effects absorb time-invariant firm heterogeneity; industry-year fixed effects absorb industry-specific shocks; some specifications add province-year fixed effects to soak up place-specific shocks from the broader 2008–2009 stimulus.
- **Interpretation of γ.** The coefficient measures the post-2009 relative change in domestic firms' investment compared with foreign firms already enjoying the tax deduction.

3.7 Production-function estimation and system GMM

The structural step starts from

$$\pi_{it} = \theta k_{it} + (1-\theta)a_{it}, \tag{14}$$

with $a_{it} = b_t + \omega_i + \varepsilon_{it}$. Using revenue as a proxy for profits and assuming $\varepsilon_{it} = \rho_\varepsilon \varepsilon_{i,t-1} + e_{it}$, the paper estimates

$$r_{it} = \rho_\varepsilon r_{i,t-1} + \theta k_{it} - \rho_\varepsilon \theta k_{i,t-1} + b_t^* + \omega_i^* + m_{it} - \rho_\varepsilon m_{i,t-1} + (1-\theta)e_{it}, \tag{15}$$

and, in first differences,

$$\Delta r_{it} = \rho_\varepsilon \Delta r_{i,t-1} + \theta \Delta k_{it} - \rho_\varepsilon \theta \Delta k_{i,t-1} + \Delta b_t^* + \Delta m_{it} - \rho_\varepsilon \Delta m_{i,t-1} + (1-\theta)\Delta e_{it}. \tag{16}$$

- **Why system GMM.** Capital is endogenous and potentially measured with error, so the paper uses lagged levels and differences as instruments in the Blundell-Bond system-GMM framework.
- **Key estimates.** The curvature parameter is $\theta = 0.734$, implying a markup around 1.224 in the authors' calibration. The transitory-shock persistence is $\rho_\varepsilon = 0.860$.

3.8 Method of simulated moments

Let $\phi = \{\delta, \gamma, \bar{\xi}\}$ denote economic depreciation, convex costs, and the upper bound of the fixed-cost distribution. The MSM estimator solves

$$g(\phi) = [\hat{m} - m(\phi)]' W [\hat{m} - m(\phi)]. \tag{17}$$

- **Moment set m_A.** Mean investment rate, its standard deviation, shares of firms with investment rates below 10%, 20%, and 30%, and the one-year autocorrelation of investment.
- **Moment set m_B.** The reduced-form DID effects of the reform on the investment rate and on the fraction of firms investing.
- **Interpretation.** Using m_B is crucial because the reform directly reduces partial irreversibility, which helps separate the role of partial irreversibility from fixed costs even though the tax data do not observe equipment sales.

4 Identification and research design (what makes the estimates credible)

4.1 Baseline identification idea

- The reform creates quasi-experimental variation because domestic firms gain equipment-VAT deductibility in 2009, while foreign firms in encouraged industries already had that benefit.
- The causal claim is therefore a treatment-control comparison between domestic and foreign firms around 2009.
- The identifying assumption is standard parallel trends: absent the reform, domestic and foreign firms would have had similar trends in equipment investment.
- There is **no instrument**. The design is difference-in-differences, not IV.

4.2 Why the control group is plausible

- Event-study graphs show that domestic and foreign firms have very similar pre-trends in both the extensive margin and the investment rate from 2005 to 2008.
- The 2008 corporate-income-tax reform changed statutory rates but had almost no effect on the TUCC gap between foreign and domestic firms, which helps isolate the 2009 VAT reform.
- The paper controls for industry-year shocks and, in preferred specifications, province-year shocks, which absorb differential exposure to the broader 4-trillion-yuan stimulus across industries and places.

4.3 Main threats and how the paper addresses them

- **Different observable firm characteristics.** The paper reweights the sample using inverse-probability weights and gets very similar estimates.
- **Export exposure and the global crisis.** Restricting to non-exporters leaves the estimates essentially intact.
- **State-directed credit in the 2008 stimulus.** Results remain when excluding SOEs and listed firms and when focusing on sectors weakly linked to infrastructure-stimulus industries.
- **Other tax changes.** Adding controls for firms' effective or statutory CIT rates does not change the core estimates.
- **Measurement choices.** Results are robust across balanced and unbalanced panels, alternative foreign control groups, log investment, and the inverse hyperbolic sine of investment.

4.4 Placebos and triple differences

- **Pilot-firm placebo.** Domestic and foreign firms that were already treated in the 2004 pilot program show no differential response in 2009.

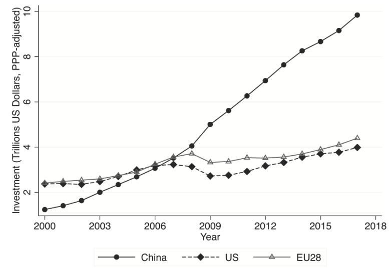

FIGURE 1

Cross-country comparison of investment

Notes: This figure displays investment in the United States, the European Union, and China from 2000 to 2017 as reported by the OECD. The figure shows that investment in China has long surpassed investment levels in the US and the European Union and that it has increased drastically since 2000.

- **Structures placebo.** The reform did not change the tax treatment of major structures, and the paper finds essentially no intensive-margin effect there.
- **Triple differences.** Using pilot firms or investment in structures as additional controls yields treatment estimates similar to the baseline DD results.
- **Takeaway.** The placebo and triple-difference results sharply reduce the risk that the main findings are driven by coincident domestic/foreign shocks rather than by the VAT reform.

5 Tables and figures

5.1 Figure 1: Cross-country comparison of investment

Read:

- China's aggregate investment rises dramatically after 2000 and overtakes both the U.S. and the EU.
- The figure motivates why understanding investment policy in China is quantitatively important.
- It also frames the reform as macro-relevant, not a niche tax change.

5.2 Figures 2 and 3: Static intuition for partial irreversibility and fixed costs

Read:

- Figure 2 shows how a purchase-resale wedge creates an inaction band in productivity space; firms only buy when productivity is high enough and only sell when productivity is low enough.
- Figure 3 shows that fixed costs create inaction by making small adjustments unprofitable even when frictionless optimal capital differs from current capital.
- These figures are the paper's conceptual core: tax policy matters because it can change the width of the inaction region.

5.3 Figure 4: Policy functions under alternative reforms

Read:

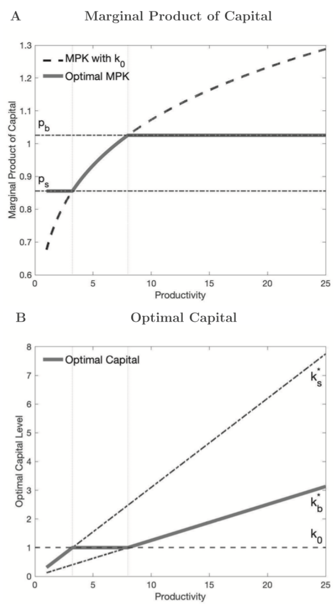

FIGURE 2

Effects of partial irreversibility

Notes: These figures plot the marginal product of capital and the optimal capital level against productivity in a simple static model with only partial irreversibility; i.e. the resale price (p^s) is smaller than the purchase price (p^b). (A) The marginal product of capital (MPK) against productivity. The dashed line corresponds to the MPK at initial capital stock k_0. The upper horizontal line indicates the purchase price p^b. The lower horizontal line indicates the resale price p^s. The thick solid line indicates the MPK at associated optimal capital levels. In (B), the thick solid line plots the optimal capital level against productivity.

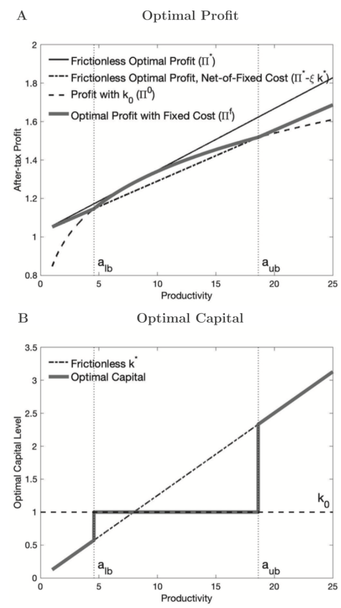

FIGURE 3

Effects of fixed costs

Notes: These figures plot the optimal profit and capital against productivity in a simple static model with fixed costs only. In (A), the solid line indicates the optimal profit without any frictions (Π^*). The dot-dashed line indicates the optimal profit net of fixed cost ($\tilde{\Pi}^* = \Pi^* - \xi k^*$). The dashed line indicates the profit evaluated at the initial capital level (Π^0). The thick solid line indicates the upper envelope of $\tilde{\Pi}^*$ and Π^0, which is the optimal profit in the presence of fixed costs. In (B), the dot-dashed line indicates the frictionless optimal capital level (k^*) against productivity. The dashed line indicates the initial capital k_0. The thick solid line indicates the optimal capital level taking into account the fixed costs.

- The authors compare a VAT cut, a CIT cut, and bonus depreciation chosen to have similar effects on the tax term of the user cost.
- The VAT cut noticeably shrinks the inaction region, whereas the CIT cut and bonus depreciation mainly shift the policy function with smaller effects on the extensive margin.
- The figure previews the main policy message: user-cost-equivalent reforms need not be investment-equivalent reforms.

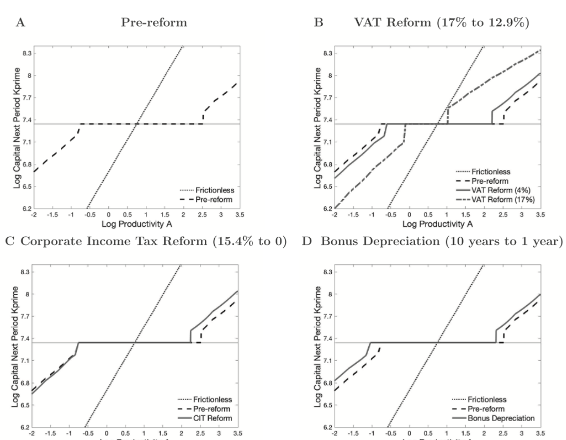

FIGURE 4

Effects of tax policy and investment frictions on policy functions

Notes: These figures display the policy functions against productivity in a dynamic investment model before and after the VAT reform, a corporate income tax reform, and the introduction of bonus depreciation, respectively. The three reforms generate the same reduction in the tax component of the user cost of capital. In (A), the dotted straight line indicates the optimal policy (the logarithm of the capital stock in the next period as a function of productivity) in the frictionless case. The dashed line indicates the optimal policy before the reform in the presence of all investment frictions—convex costs, fixed costs, and partial irreversibility. (B) The policy function (the red line) after reducing the VAT rate from 17% to 12.9%; (C) The policy function after reducing the corporate income tax rate from 15.4% to 0%; (D) The policy function after a bonus depreciation policy that fully accelerates the timing of depreciation deductions.

TABLE 1

VAT reform and investment costs: example of a 1,000 RMB equipment purchase

	Pre-reform	Post-reform	Change
VAT-included cost	1,170	1,170	
Deductible from VAT	**0**	**170**	**+170**
Book value	1,170	1,000	−170
PV of total depreciation	948.6	810.8	−137.8
Deductible from corporate income tax	237.2	202.7	−34.5
After-tax cost of investment	**932.8**	**797.3**	**−135.5**

Notes: This calculation assumes a discount rate of 5% and a marginal corporate income tax rate of 25%. According to Chinese accounting standards, the book value of the asset would be depreciated over 10 years using a straight-line depreciation method. This calculation assumes a zero salvage value.

5.4 Table 1: After-tax cost of a 1,000 RMB equipment purchase

Read:

- Pre-reform after-tax cost: 932.8 RMB. Post-reform after-tax cost: 797.3 RMB.
- The reform lowers the after-tax price by 135.5 RMB, about 15%.
- The table makes the policy shock very concrete and shows that the direct VAT credit effect dominates the loss of future depreciation deductions.

5.5 Table 2 and Figure 5: Descriptives and lumpy investment patterns

TABLE 2
Summary statistics

	All firms			Domestic firms			Foreign firms		
	Mean	SD	*N*	Mean	SD	*N*	Mean	SD	*N*
Equipment investment (million RMB)									
Investment	3.15	12.37	221,069	2.61	11.02	202,155	9.01	21.27	18,914
Investment rate	0.10	0.19	215,813	0.10	0.19	197,050	0.10	0.16	18,763
Log investment	6.50	2.31	118,913	6.36	2.27	104,929	7.54	2.36	13,984
Other characteristics (million RMB)									
Total investment	4.70	17.27	258,736	4.00	15.71	234,475	11.40	27.33	24,261
Fixed assets	33.77	100.32	310,003	28.85	91.57	282,582	84.48	156.65	27,421
Sales	133.80	390.83	314,595	114.79	351.52	287,033	331.77	643.16	27,562
Cash inflow	126.74	384.06	283,694	106.45	344.23	257,280	324.38	622.06	26,414
Debt	85.13	241.28	313,074	77.58	228.86	285,570	163.49	334.81	27,504

Notes: This table presents summary statistics of equipment investment and other variables from tax data that are used in the analysis. Investment is reported in million RMB and deflated by the national price index of equipment investment. The investment rate is defined as the ratio of investment to the capital stock measured in terms of the book value of net fixed assets, unconditional on investing. Total investment includes investments in equipment, buildings and structures, and other productive capital. Fixed assets are measured in terms of the book value, deflated by the national price index of fixed-asset investment. Sales are the total sales including domestic and export sales. Cash flow is the business cash inflow from the cash-flow statement. Debt is the total debt at the end of the year. Variables are winsorized at the 1% level.

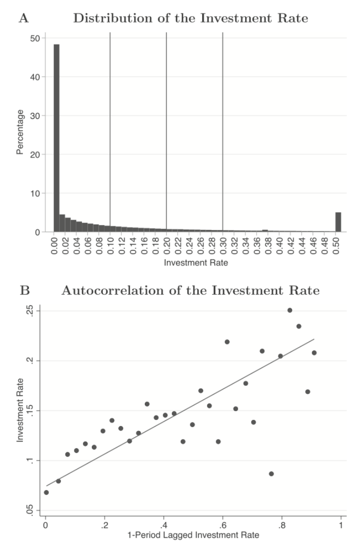

FIGURE 5
Distribution and autocorrelation of the investment rate

Notes: These figures display notable features of the investment in our tax data. (A) The distribution of the investment rate of domestic firms before the reform. We winsorize the investment rate at the top 5%. (B) The investment rate against the one-period-lagged investment rate. We group the lagged investment rate into equally sized bins from 0 to 1 and then calculate the average investment rate for each bin. The red line is the OLS linear fit line.

Read:

- Table 2 shows average equipment-investment rates of about 10% for both domestic and foreign firms, despite large level differences in firm size.
- Figure 5 shows a huge mass at zero investment and a fat right tail of large spikes, with a one-year investment autocorrelation around 0.20.
- This is exactly the evidence that motivates a model with both nonconvex and convex adjustment costs.

5.6 Figure 6 and Table 3: Baseline reduced-form evidence

Read:

- Figure 6 shows flat pre-trends through 2008 and then a discrete post-2009 jump for domestic firms relative to foreign firms.

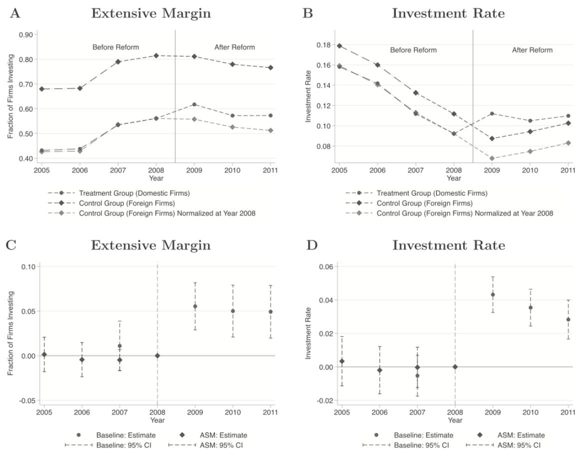

FIGURE 6

Reduced-form effects of China's 2009 VAT reform

Notes: These figures show the effects of the VAT reform on the investment of domestic and foreign firms. (A) The fraction of firms investing in equipment in each year for domestic firms (the treatment group) and foreign firms (the control group). The dashed line with circular markers represents investment of domestic firms, the upper dashed line with diamond markers represents investment of foreign firms, and the lower dashed line with diamond markers represents the investment of foreign firms that is normalized to that of domestic firms in 2008. To construct the figure, we first use tax data to calculate the fraction of firms investing in equipment for each year from 2007 to 2011 for domestic and foreign firms. In addition, we complement the tax data with Chinese Annual Survey of Manufacturing (ASM) data. That is, we merge the 2005–06 ASM data with the tax data and calculate the fraction of firms with positive total investment for each year for domestic and foreign firms, which allows us to extend the pre-reform investment to 2005. For easier comparison, we set 2008 as the base year and align the fraction of firms investing in equipment for foreign firms to that for domestic firms in 2008, as showed in the lower dashed line with diamond markers. Similarly, (B) plots the average investment rate of equipment for each year for domestic and foreign firms. (C) plots the differences in the fraction of firms investing between domestic and foreign firms. The circular dots correspond to the estimates from the tax data from 2007 to 2011; The diamond dots correspond to the estimates from the extended data with ASM. Again, for easier comparison, we set 2008 as the base year. That is, we subtract the difference in the fraction of firms investing between domestic and foreign firms in 2008 from the difference of each year. Similarly, (D) plots the differences in average investment rate in equipment between domestic and foreign firms.

TABLE 3

Estimates of difference-in-difference regressions

		Extensive margin: % firms			Intensive margin: investment rate		
		(1)	(2)	(3)	(4)	(5)	(6)
A. Main results							
Domestic	Post	0.058***	0.044***	0.046***	0.036***	0.037***	0.038***
		(0.010)	(0.010)	(0.010)	(0.004)	(0.004)	(0.004)
N		86,870	86,870	86,870	81,270	81,270	81,270
Industry	Year FE	Y		Y	Y		Y
Province	Year FE		Y	Y		Y	Y
B. Robustness checks: sample selection							
		IPW	Unbalanced	All foreign	IPW	Unbalanced	All foreign
Domestic	Post	0.060***	0.053***	0.045***	0.033***	0.044***	0.039***
		(0.021)	(0.008)	(0.006)	(0.008)	(0.003)	(0.003)
N		82,785	221,069	107,255	79,195	215,813	100,980
Industry	Year FE	Y	Y	Y	Y	Y	Y
Province	Year FE	Y	Y	Y	Y	Y	Y
C. Robustness checks: different samples							
		Non-exporters	Non-SOE	Non-public	Non-exporters	Non-SOE	Non-public
Domestic	Post	0.067***	0.046***	0.046***	0.037***	0.037***	0.038***
		(0.020)	(0.010)	(0.010)	(0.009)	(0.004)	(0.004)
N		61,195	83,653	85,360	56,445	78,238	79,855
Industry	Year FE	Y	Y	Y	Y	Y	Y
Province	Year FE	Y	Y	Y	Y	Y	Y
D. Additional firm-level controls							
		CF	Firm controls	CIT	CF	Firm controls	CIT
Domestic	Post	0.043***	0.048***	0.052***	0.038***	0.035***	0.036***
		(0.010)	(0.010)	(0.012)	(0.004)	(0.004)	(0.005)
N		83,418	86,284	86,870	79,547	80,823	81,270
Industry	Year FE	Y	Y	Y	Y	Y	Y
Province	Year FE	Y	Y	Y	Y	Y	Y
E. Alternative investment measures							
		Log investment			IHS investment		
Domestic	Post	0.449***	0.404***	0.414***	0.711***	0.624***	0.646***
		(0.052)	(0.054)	(0.054)	(0.083)	(0.086)	(0.087)
N		20,720	20,720	20,720	86,870	86,870	86,870
Industry	Year FE	Y		Y	Y		Y
Province	Year FE		Y	Y		Y	Y

Notes: This table uses tax data to estimate difference-in-difference regressions of the form:

$$Y_{it} = \gamma G_i \times \text{Post}_t + \mu_i + \delta_{jt} + X'_{it}\beta + \varepsilon_{ijt},$$

where Y_{it} is equipment investment, G_i is the treatment indicator set to 1 for domestic firms and 0 for foreign firms, and Post_t is the post-reform indicator set to 1 for years since 2009. μ_i is the firm fixed effect. Panel A reports the baseline results. The dependent variable for Columns (1)–(3) is a dummy variable set to 1 if a firm makes an investment; the dependent variable for Columns (4)–(6) is firm's investment rate. Columns (1) and (4) control for industry-year fixed effects. Columns (2) and (5) control for province-year fixed effects. Columns (3) and (6) include both fixed effects. Panel B reports robustness checks: Column (1) weights observations by the inverse probability weighting (IPW); Column (2) uses an unbalanced panel; Column (3) uses all foreign firms as the control group. Panel C reports another set of robustness checks: Column (1) exclude exporters; Column (2) exclude state-owned firms (SOEs); Column (3) excludes public firms. Panel D augments the regression with additional controls. Column (1) controls for firms' net cash flow scaled by the capital stock. Column (2) adds quadratics in sales, profit margin, and age. Column (3) adds the statutory corporate income tax rate. Panel E runs the baseline specification with the log and inverse hyperbolic sine of investment as dependent variables. All regressions include firm fixed effects. Standard errors are clustered at the firm level.

- Table 3 quantifies the jump: about 4.4 to 5.8 percentage points on the extensive margin and about 3.6 to 3.8 percentage points on the intensive margin, depending on fixed effects and controls.
- Relative to a base investment rate of 10%, the preferred intensive-margin estimate is a 36% increase.
- The preferred specifications include both industry-year and province-year fixed effects, with coefficients around 0.046 for participation and 0.038 for the investment rate.

5.7 Tables 4 and 5: Placebo tests and triple differences

Read:

- In pilot firms, the post-2009 domestic-foreign difference is near zero: about −0.010 on the participation margin and 0.017 on the investment rate, both imprecise.
- For structures, the intensive-margin placebo is essentially zero (about −0.002).
- Triple-difference estimates remain close to the baseline: around 0.030 on the extensive margin using pilot firms and around 0.033 on the intensive margin using equipment-versus-structures comparisons.
- These exhibits are central for credibility because they show the result is not just a generic domestic-firm shock in 2009.

TABLE 4

Estimates of placebo tests

		Extensive margin: % firms investing			Intensive margin: investment rate		
		(1) Baseline	(2) Pilot	(3) Structures	(4) Baseline	(5) Pilot	(6) Structures
Domestic	Post	0.046***	−0.010	0.023**	0.038***	0.017	−0.002
		(0.010)	(0.032)	(0.011)	(0.004)	(0.013)	(0.003)
N		86870	13932	81270	81270	12340	81270
Industry	Year FE	Y	Y	Y	Y	Y	Y
Province	Year FE	Y	Y	Y	Y	Y	Y

Notes: This table uses tax data to estimate difference-in-difference regressions of the form:

$$Y_{it} = \gamma G_i \times \text{Post}_t + \mu_i + \delta_{jt} + \eta_{st} + \varepsilon_{it},$$

where Y_{it} is a measure of investment, G_i is the treatment indicator set to 1 for domestic firms and 0 for foreign firms, and Post_t is the post-reform indicator set to 1 for years since 2009. μ_i is the firm fixed effect. δ_{jt} and η_{st} are industry-year and province-year fixed effects, respectively. Column (1) and (4) report the baseline results, where the non-pilot domestic firms are the treatment group and foreign firms with preferential treatment are the control group. Column (2) and (5) use domestic firms and foreign firms in the pilot program as the treatment and control groups, respectively. Columns (3) and (6) use investment in buildings/structures as dependent variables, and the treatment and control groups are the same as those in the baseline analysis. The dependent variable for Columns (1) and (2) is a dummy variable set to 1 if a firm makes an investment in equipment; the dependent variable for Column (3) is a dummy variable set to 1 if a firm makes an investment in structures with investment rate larger than %. The dependent variable is firm's equipment investment rate for Columns (4) and (5) and is firm's structures investment rate for Column (6). All regressions include firm fixed effects as well as industry-year and province-year fixed effects. Standard errors are clustered at the firm level.

TABLE 5

Estimates of triple-differences regressions

			Extensive margin: % firms investing			Intensive margin: investment rate		
			(1) Baseline	(2) Pilot	(3) Structures	(4) Baseline	(5) Pilot	(6) Structures
Domestic	Post		0.046***	−0.036	0.028**	0.038***	0.011	0.002
			(0.010)	(0.030)	(0.011)	(0.004)	(0.012)	(0.003)
Domestic	Post	Non-Pilot		0.080**			0.030**	
				(0.032)			(0.013)	
Domestic	Post	Equipment			0.036**			0.033***
					(0.015)			(0.005)
N			86,870	100,040	162,540	81,270	92,875	162,540
Industry	Year FE		Y	Y	Y	Y	Y	Y
Province	Year FE		Y	Y	Y	Y	Y	Y

Notes: This table uses tax data to estimate triple-differences regressions. For comparison, Columns (1) and (4) report the baseline estimates from difference-in-difference regressions. Columns (2) and (5) use firms in the pilot program as an additional control group and estimate the triple-differences regression of the form:

$$Y_{it} = \gamma_1 G_i \times \text{Post}_t + \gamma_2 P_i \times \text{Post}_t + \gamma_3 G_i \times P_i \times \text{Post}_t + \mu_i + \delta_{jt} + \eta_{st} + \varepsilon_{it},$$

where Y_{it} is equipment investment, P_i is an indicator set to 1 for pilot firms and 0 for non-pilot firms, G_i is the treatment indicator set to 1 for domestic firms and 0 for foreign firms, and Post_t is the post-reform indicator set to 1 for years since 2009. μ_i is the firm fixed effect. δ_{jt} and η_{st} are industry-year and province-year fixed effects, respectively. Columns (3) and (6) use investment in structures as an additional control group and estimate the triple-differences regression of the form:

$$Y_{kit} = \gamma_1 G_i \times \text{Post}_t + \gamma_2 A_k \times \text{Post}_t + \gamma_3 G_i \times A_k \times \text{Post}_t + \gamma_4 G_i \times A_k + \mu_i + \delta_{jt} + \eta_{st} + \varepsilon_{kit},$$

where k denotes the investment type, i.e. equipment and structures. A_k is an indicator set to 1 for investment in equipment and 0 for investment in structures. The other variables are defined the same as above. The dependent variable for Columns (1)–(3) is a dummy variable set to 1 if a firm makes an investment; the dependent variable for Columns (4)–(6) is the firm's investment rate. All regressions include firm fixed effects as well as industry-year and province-year fixed effects. Standard errors are clustered at the firm level.

5.8 Table 6: Investment spikes

Read:

- The probability of an investment spike (investment rate ≥ 0.2) rises by about 7.0 to 7.3 percentage points.
- The spike component of investment rises by about 3.1 to 3.5 percentage points, whereas the non-spike component rises only by about 0.3 to 0.5 percentage points.
- Quantitatively, investment spikes explain roughly 86% to 92% of the overall increase in the investment rate.
- This is the cleanest evidence that the extensive margin is doing the heavy lifting.

TABLE 6

Estimates of difference-in-difference regressions: investment spikes

		Extensive margin:			Intensive Margin:					
		% Firms investing with $IK \geq 0.2$			Spike investment rate			Non-spike investment rate		
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Domestic	Post	0.070***	0.071***	0.073***	0.031***	0.034***	0.035***	0.005***	0.003**	0.003**
		(0.009)	(0.009)	(0.010)	(0.004)	(0.004)	(0.004)	(0.001)	(0.001)	(0.001)
N		81,270	81,270	81,270	81,270	81,270	81,270	81,270	81,270	81,270
Industry	Year FE	Y		Y	Y		Y	Y		Y
Province	Year FE		Y	Y		Y	Y		Y	Y

Notes: This table uses tax data to estimate difference-in-difference regressions of the form:

$$Y_{it} = \gamma G_i \times \text{Post}_t + \mu_i + \delta_{jt} + \eta_{st} + \varepsilon_{it},$$

where Y_{it} is a measure regarding investment spikes, G_i is the treatment indicator set to 1 for domestic firms and 0 for foreign firms, and Post_t is the post-reform indicator set to 1 for years since 2009. μ_i is the firm fixed effect. δ_{jt} and η_{st} are industry-year and province-year fixed effects, respectively. The dependent variable for Columns (1)–(3) is a dummy variable set to 1 if the investment rate is larger than 0.2, i.e. $D_{it}^{\text{spike}} = \mathbb{1}\{IK_{it} \geq 0.2\}$, where IK_{it} is the investment rate of firm i at time t. The dependent variable for Columns (4)–(6) is the spike investment rate, defined by $IK_{it}^{\text{spike}} = IK_{it} \times \mathbb{1}\{IK_{it} \geq 0.2\}$; the dependent variable for Columns (7)–(9) is the non-spike investment rate, defined by $IK_{it}^{\text{non-spike}} = IK_{it} \times \mathbb{1}\{IK_{it} < 0.2\}$. All regressions include firm fixed effects. Standard errors are clustered at the firm level.

TABLE 7

Summary of assigned and estimated parameters

Description		*Value*(*S.E*)
Panel A. Fixed parameters		
Discount factor	β	0.950
VAT rate	ν	0.170
CIT rate	τ	0.154
PV depreciation schedule	z	0.803
Panel B. Parameters estimated via system GMM		
Profit curvature	θ	0.734(0.031)
Persistence firm transitory shocks	ρ_ε	0.860(0.011)
SD firm transitory shocks	σ_ε	0.529(0.005)
SD firm permanent shocks	σ_ω	0.854(0.007)
Persistence aggregate shocks	ρ_b	0.009(0.152)
SD aggregate shocks	σ_b	0.010(0.001)
Panel C. Parameters estimated by MSM		
Convex cost	γ	1.434(0.066)
Upper bound of fixed cost	$\bar{\xi}$	0.118(0.004)
Economic depreciation rate	δ	0.071(0.001)

Notes: This table summarizes the parameters from Section 6. Panel A displays the parameters we set (i.e. those not estimated) to simulate the model. Specifically, we set tax parameters to their empirical counterparts. Panel B summarizes the estimated parameters from the first-stage production function estimation and productivity decomposition. In particular, we estimate the profit curvature (θ) and the persistence of firm transitory shocks (ρ_ε) using system GMM. Standard errors are reported in parentheses. The rest of the parameters in Panel B are the results of the productivity decomposition (see Section 6.1). The standard errors of those parameters (i.e. σ_ε, σ_ω, ρ_b, and σ_b) are calculated from 100 bootstrap samples. Panel C displays the estimated adjustment frictions and depreciation rate using the method of simulated moments (MSM) (see Section 6.2). Standard errors of those parameters (i.e. γ, $\bar{\xi}$, and δ) are calculated from 100 bootstrap samples.

5.9 Table 7: Parameters from system GMM and MSM

Read:

- Fixed tax parameters used in simulation are $\beta = 0.950$, $\nu = 0.170$, $\tau = 0.154$, and $z = 0.803$.
- Estimated profit and shock parameters are $\theta = 0.734$, $\rho_\varepsilon = 0.860$, $\sigma_\varepsilon = 0.529$, $\sigma_\omega = 0.854$, $\rho_b = 0.009$, and $\sigma_b = 0.010$.

TABLE 8

Structural estimation and moments

Model	Parameters			Moments							
	γ	$\bar{\xi}$	δ	Avg i	Share $i<0.1$	Share $i<0.2$	Share $i<0.3$	Corr(i,i_{-1})	SD i	DID. Ext	DID. Int
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Data				0.102	0.720	0.834	0.891	0.200	0.186	0.046	0.038
				(0.001)	(0.003)	(0.002)	(0.002)	(0.013)	(0.002)	(0.010)	(0.004)
Fixed	0.427	0.074	0.087	0.113	0.838	0.838	0.840	0.273	0.249	0.078	0.086
(A)	(0.141)	(0.012)	(0.001)								
Uniform	1.594	0.118	0.071	0.080	0.742	0.793	0.879	0.198	0.138	0.027	0.026
(A)	(0.090)	(0.003)	(0.001)								
Uniform	1.434	0.118	0.071	0.081	0.747	0.795	0.875	0.208	0.140	0.036	0.030
(A+B)	(0.066)	(0.004)	(0.001)								
Average fixed cost (conditional on investing): 0.024											

Notes: This table displays estimates of the convex adjustment cost γ, upper bound of fixed cost $\bar{\xi}$, and economic depreciation rate δ using the method of simulated moments. Columns (4)–(11) show the simulated moments: (1) Set A includes pre-reform static moments, namely, the average investment rate; the fraction of firms with an investment rate smaller than 0.1, 0.2, and 0.3, the one-period serial correlation of the investment rate, and the standard deviation of investment rate. (2) Set B includes difference-in-difference estimates at the extensive and intensive margins. In particular, we simulate 10,000 firms over 200 periods, with the reform taking place at the 100th period. We use the last 20 periods before the reform to calculate the pre-reform static moments. Meanwhile, we simulate another counterfactual economy where the reform does not take place using the same productivity shocks. The difference-in-difference moments are calculated by taking the difference between the two simulated economies. The first row reports the data moments with standard errors in parentheses calculated using 100 bootstrap samples. The second row assumes the fixed cost is non-random and uses the pre-reform static moments for estimation. The third and the fourth rows (i.e. Uniform (A) and Uniform (A+B)) assume that the fixed cost is independently and identically distributed across firms and over time, following a uniform distribution over $[0,\bar{\xi}]$. The third row uses pre-reform static moments for estimation. The fourth row uses both pre-reform static moments and the difference-in-difference moments for estimation. In the last row, using model Uniform (A+B), we report the average fixed cost for firms making a positive investment. The standard errors of the estimates are reported in parentheses below the point estimates.

- Preferred adjustment-cost estimates are $\gamma = 1.434$, $\bar{\xi} = 0.118$, and $\delta = 0.071$.
- These values imply moderate convex costs and economically meaningful fixed disruption costs.

5.10 Table 8: Structural fit to moments

Read:

- The fixed-cost-only mass-point model fits the average investment rate and spike share poorly and massively overpredicts the reform response.
- The preferred uniform-fixed-cost model that matches both pre-reform moments and reform moments reproduces the data closely: average investment rate about 0.081, serial correlation about 0.208, DID extensive margin about 0.036, and DID intensive margin about 0.030.
- The model's average fixed cost conditional on investing is about 0.024, or 2.4% of desired capital.
- The exhibit matters because it shows the model is disciplined by both stationary lumpy-investment moments and quasi-experimental causal evidence.

5.11 Figure 7 and Table 9: Simulating alternative tax reforms

Read:

- Figure 7 shows that for a given revenue loss, VAT cuts and investment tax credits raise investment much more than CIT cuts.
- Table 9 reports the magnitudes over a 10-year window. A full VAT cut raises aggregate investment by 43.4%, the fraction of investing firms by 9.8%, and firm value by 11.4%, with a tax-revenue loss of 27.9%.
- A CIT cut from 15.4% to 10% raises investment by only 14.7% despite a revenue loss of 19.1%. Bonus depreciation raises investment by 9.9%. A TCJA-style combination raises investment by 21.7%.
- A 17% investment tax credit looks very similar to the VAT reform: aggregate investment rises

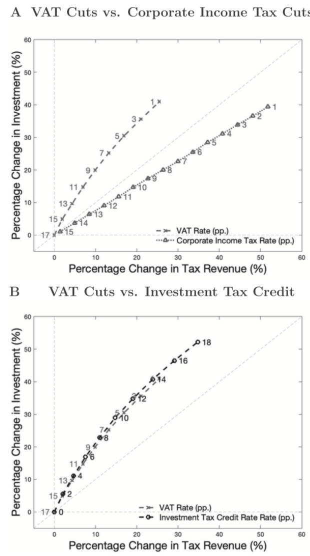

FIGURE 7

Simulating alternative tax reforms: elasticity of investment to tax revenue

Notes: These figures plot the simulated percentage change in aggregate investment to the percentage loss in tax revenue at different rates of VAT cut, corporate income tax cut, and investment tax credit policies. For each tax rate, we solve the model, simulate investment and tax revenue, and calculate the corresponding changes in each outcome. (A) plots the percentage change in aggregate investment against the percentage change in tax revenue. The dashed line with X markers corresponds to VAT cuts from 17% to different rates. The dotted line with triangle markers corresponds to corporate income tax cuts from 15.4% to different rates. Similarly, (B) compares the percentage change in tax revenue from VAT cuts with that from an investment tax credit.

TABLE 9

Simulating tax reforms

	Baseline 17%	Corporate tax cut 15.4% to 10%	Bonus depreciation	Tax cuts and jobs act: (2)+(3)	Investment tax credit 17%
Percentage change in	(1)	(2)	(3)	(4)	(5)
Aggregate investment	0.434	0.147	0.099	0.217	0.495
Fraction of firms investing	0.098	0.062	0.032	0.084	0.101
Tax revenue	−0.279	−0.191	−0.131	−0.286	−0.319
Firm value	0.114	0.103	0.017	0.116	0.132
Ratios of					
Investment to tax revenue	1.559	0.769	0.753	0.757	1.553
Firm value to tax revenue	0.410	0.540	0.134	0.407	0.415

Notes: This table displays the simulated responses to five scenarios: Column (1) considers a reduction of the VAT rate from 17% to 0%, i.e. our baseline reform; Column (2) considers a reduction of the effective corporate income tax rate from 15.4% to 10%; Column (3) considers a policy that allows firms to fully depreciate capital expenses immediately, i.e. bonus depreciation; Column (4) considers a combination of a corporate income tax cut and bonus depreciation, i.e. a version of the Tax Cuts and Jobs Act (TCJA); and Column (5) considers a policy granting a 17% investment tax credit. We report percentage (%) changes in the outcomes of interest. For instance, our baseline 17% VAT cut increases aggregate investment by 43% over a 10-year window.

by 49.5% with an investment-to-tax-revenue ratio of about 1.553, almost identical to the VAT cut's 1.559.

- The paper's headline policy conclusion comes from this exhibit: policies that reduce inaction are fiscally more effective than policies that broadly lower tax rates for all firms, including inframarginal non-investors.

6 Short summary

- **Conclusion.** China's 2009 VAT reform substantially increased equipment investment, and the increase came mainly from firms switching from inaction to large investment spikes.
- **Intuition.** The reform not only lowered the user cost of capital; it also reduced partial irreversibility by narrowing the gap between the purchase and resale prices of capital.
- **Empirical lesson.** To understand the effects of tax reform, one must model and estimate both the intensive and extensive margins of investment.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Tax policy can directly alter the lumpiness of investment by changing the frictions that determine whether firms invest at all.
- **Magnitude.** The 2009 Chinese VAT reform lowered the after-tax cost of investment by about 15%, raised the probability of investing by about 5 percentage points, and increased the investment rate by about 36% relative to the pre-reform mean.

- **Credibility.** Pre-trends are flat, a wide set of sample and control variations leave the estimates unchanged, and pilot-firm / structures placebos plus triple differences support the identification strategy.
- **Mechanism evidence.** The reform strongly raises the frequency of investment spikes, exactly as predicted by a model in which VAT distortions widen the inaction region through partial irreversibility.
- **Quantitative implication.** Structural estimates imply that VAT cuts and investment tax credits are more effective per unit of revenue loss than corporate-income-tax cuts or bonus depreciation because they disproportionately move firms out of inaction.

7.2 Economic intuition

- Firms with lumpy investment do not respond continuously to small tax changes. They sit in an inaction band until productivity is high enough to justify paying the nonconvex adjustment cost.
- A tax on purchases of new capital is especially distortionary because it raises the purchase price without equally raising the resale price. That widens the inaction region and suppresses spikes.
- Removing that wedge has first-order extensive-margin effects. In contrast, a broad income-tax cut also benefits many inframarginal firms whose investment does not change, so its fiscal bang-for-buck is smaller.

7.3 Takeaway

This paper's punchline is simple but important: once investment is lumpy, "lowering the user cost" is not enough to characterize tax policy. The specific tax instrument matters because it changes the structure of adjustment frictions. In China's case, eliminating the VAT distortion on equipment investment triggers many firms to move from inaction to large investment spikes, and that is why the policy is so powerful relative to alternative tax reforms.